

MA1513 Weekly Self Practice Set 5

1. Let $y_1(t)$, $y_2(t)$ and $y_3(t)$ be three functions in t such that

$$\begin{aligned}y_1' &= 5y_1 - y_2 - y_3 \\y_2' &= -y_1 + 5y_2 - y_3 \ . \\y_3' &= -y_1 - y_2 + 5y_3\end{aligned}$$

- (a) Write down the matrix form of the above system of differential equations.
- (b) Use MATLAB command to find the eigenvalues and eigenvectors of the underlying matrix in (a), and use them to write down a fundamental set of solutions for the above SDE.

(To get nice values for eigenvectors, use the MATLAB symbolic command:
`>> [P D]=eig(sym(A)).`)

- (c) Write down the general solution of the SDE.
- (d) Given the initial conditions $y_1(0) = 3, y_2(0) = 3, y_3(0) = 0$, find the particular solution of the initial value problem for the above SDE. Show your working.